

BIOGRAPHICAL SKETCH

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NAME Andrews, Steven S.	POSITION TITLE Research fellow		
eRA COMMONS USER NAME			
EDUCATION/TRAINING <i>(Begin with baccalaureate or other initial professional education, such as nursing, and include postdoctoral training.)</i>			
INSTITUTION AND LOCATION	DEGREE <i>(if applicable)</i>	YEAR(s)	FIELD OF STUDY
Dartmouth College	B.A.	1991	Chemistry
Stanford University	Ph.D.	2001	Chemistry
Stanford University	Ph.D. minor	2001	Physics

A. Positions and Honors.**Positions and Employment**

1991 Undergraduate Research, Dartmouth College, Hanover, NH
 1991 Research Assistant, Dartmouth College, Hanover, NH
 1991-1992 Intern for Meteorology and Icing Research, Mount Washington Observatory, Gorham, NH
 1993-1994 Research Assistant, Woods Hole Oceanographic Institution, Woods Hole, MA
 2001-2003 Research Associate, University of Cambridge, Cambridge, U.K.
 2003-2007 Research Associate, Lawrence Berkeley National Laboratory, Berkeley, CA
 2006-2007 Consultant, Molecular Sciences Institute, Berkeley, CA
 2007 Visiting Scientist, National Centre for Biological Sciences, Bangalore, India

Other Experience and Professional Memberships

1992 Teacher, Barrier Island Environmental Education Center, Seabrook Island, South Carolina
 1994-1995 Teaching Assistant, Stanford University, Stanford, CA
 1995 Head Teaching Assistant, Stanford University, Palo Alto, CA
 1997-2001 Volunteer instructor, Stanford University Outdoor Education Program, Stanford, CA
 2004-2005 Mentor for students, Lawrence Berkeley National Laboratory, Berkeley, CA
 2006-2007 Mentor for student, University of California at Berkeley, Berkeley, CA

Honors

1988 First place Thayer Math Prize
 1991 Cum Laude with high honors in Chemistry
 1991 Haseltine Chemistry-Physics Prize
 2001 Annual Reviews Prize in Physical Chemistry
 2003-2005 Post-doc fellowship in biological Informatics, National Science Foundation

B. Selected peer-reviewed publications (in chronological order).

Lipson, J.E.G. and S.S. Andrews. "A Born-Green-Yvon Integral Equation Treatment of a Compressible Fluid." J. Chem. Phys. 96:1426-1434, 1992.

Weiss, Peter S., Steven S. Andrews, James E. Johnson, and Oliver C. Zafiriou. "Photoproduction of carbonyl sulfide in South Pacific Ocean waters as a function of irradiation wavelength." Geophys. Res. Lett. 22:215-218, 1995.

Park, Eun Sun, Steven S. Andrews, Robert B. Hu, and Steven G. Boxer. "Vibrational Stark spectroscopy in proteins: A probe and calibration for electrostatic fields." *J. Phys. Chem. B* 103: 9813-9817, 1999.

Andrews, Steven S., Sigalit Caron, and Oliver C. Zafiriou. "Photochemical oxygen consumption in marine waters: A major sink for colored dissolved organic matter?" *Limnol. Oceanogr.* 45:267277, 2000.

Andrews, Steven S. and Steven G. Boxer. "A liquid nitrogen immersion cryostat for optical measurements." *Rev. Sci. Instr.* 71:3567-3569, 2000.

Andrews, Steven S. and Steven G. Boxer. "Vibrational Stark effects of nitriles. I. Methods and experimental results." *J. Phys. Chem. A* 104:11853-11863, 2000.

Andrews, Steven S. and Steven G. Boxer. "Analysis of noise for rapid-scan and step-scan FTIR difference spectroscopy." *Appl. Spectrosc.* 55:1161-1165, 2001.

Andrews, Steven S. and Steven G. Boxer. "Vibrational Stark effects of nitriles. II. Physical origins of Stark effects from experiment and perturbation models." *J. Phys. Chem A* 106:469-477, 2002.

Xie, Huixiang, Steven S. Andrews, William R. Martin, Jared Miller, Lori Ziolkowski, Craig D. Taylor, and Oliver C. Zafiriou. "Validated methods for sampling and headspace analysis of carbon monoxide in seawater." *Marine Chem.* 77:93-108, 2002.

Zafiriou, Oliver C., Steven S. Andrews, and Wei Wang. "Concordant estimates of oceanic carbon monoxide source and sink processes in the Pacific yield a balanced global 'blue-water' CO budget." *Global Biogeochemical Cycles* 17:1015, 2003.

Treynor, Thomas P., Steven S. Andrews, and Steven G. Boxer. "Intervalence band Stark effect of the special pair radical cation in bacterial photosynthetic reaction centers." *J. Phys. Chem. B* 107:11230-11239, 2003.

Andrews, Steven S. "Using rotational averaging to calculate the bulk response of isotropic and anisotropic samples from molecular parameters." *J. Chem Ed.* 81:877-885, 2004.

Goldman, Jacki, Steven S. Andrews, and Dennis Bray. "Size and composition of membrane protein clusters predicted by Monte Carlo analysis." *Eur. Biophys. J.* 33:506-512, 2004.

Adelman, Joshua L. and Steven S. Andrews. "Intracellular pattern formation: A spatial stochastic model of bacterial division site selection proteins MinCDE." *Proc. Santa Fe Inst. Summer School*, 2004.

Andrews, Steven S. and Dennis Bray. "Stochastic simulation of chemical reactions with spatial resolution and single molecule detail." *Phys. Biol.* 1:137-151, 2004. Chosen as one of 6 Physical Biology Highlights of 2004-2005.

Lipkow, Karen, Steven S. Andrews, and Dennis Bray. "Simulated diffusion of CheYp through the cytoplasm of *E. coli*." *J. Bact.* 187:45-53, 2005.

Andrews, Steven S. "Serial rebinding of ligands to clustered receptors as exemplified by bacterial chemotaxis." *Phys. Biol.* 2:111-122, 2005.

Andrews, Steven S. and Adam P. Arkin. "Simulated niche partitioning by bacteria." *Proc. Intl. Conf. on Complex Systems*, 2005.

Andrews, Steven S. and Adam P. Arkin. "Simulating Cell Biology." Curr. Biol. 16:R523-R527, 2006.

Andrews, Steven S. "Review of System Modeling in Cellular Biology: From Concepts to Nuts and Bolts, edited by Zoltan Szallasi, Jörg Stelling, and Vipul Periwal" Quart. Rev. Biol. 82:145, 2007.

Andrews, Steven S. and Adam P. Arkin. "Systems biology: A switch for sex." Current Biol. 17:R410, 2007.

Andrews, Steven S. and Adam P. Arkin. "A mechanical explanation for cytoskeletal rings and helices in bacteria." Biophys. J., 93:1872-1884, 2007.

Andrews, Steven S., Tuan Dinh, and Adam P. Arkin. "Stochastic modeling of biochemical reaction networks." Encyclopedia of Complexity and System Science, in press, 2008.

C. Research Support

none